The Economics of Overloading: Capacity Standards, Shadow Prices, and the Viability Gap

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Abstract

Overloading — carrying more passengers or freight than the law permits — is universal, persistent, and illegal. This paper argues that it is not a failure of enforcement but a **structural consequence of the gap between the legal capacity and the economic threshold**. When the critical batch size $n^* = c_0/(p - c_v)$ exceeds the legal capacity K_L, the vehicle is economically unviable at legal load, and overloading is the only way to reach the threshold. The fine is not a deterrent; it is a **price the state sets for the right to overload**, and the observed overload rate is the shadow price of a safety standard the corridor's economics cannot afford. We derive the enforcement equilibrium, show that the state tolerates overloading because removing it would remove the service, and extend the analysis to axle-load limits, where the externality is road damage and the damage function is convex (the fourth power law). The paper's central claim is that **persistent overloading is a diagnostic, not a crime**: it identifies a corridor in which the legal capacity has been set below the economic threshold, and the correct intervention is to close the gap, not to punish the operator who is trying to survive it.

1 The Puzzle

Overloading is illegal everywhere and practised everywhere. A matatu licensed to carry fourteen passengers carries eighteen. A truck rated for thirty tonnes carries forty-five. A ferry certified for two hundred passengers carries three hundred. The fines exist, the enforcement exists, the accidents happen, and the practice continues.

The conventional explanations are familiar: weak enforcement, corrupt officials, poor safety culture, poverty. Each is partly true. None explains the *structure* of the phenomenon: overloading is concentrated in low-income corridors, in rural routes, in old vehicles, and in exactly the services that are cheapest and most needed. If overloading were simply a failure of enforcement, it would be randomly distributed. It is not. It has a pattern, and the pattern is economic.

This paper argues that overloading is the predictable consequence of a specific mismatch: **the legal capacity has been set below the economic threshold**. When that happens, the operator who obeys the law loses money on every trip, and the operator who breaks it survives. Overloading is not a moral failure. It is the market's way of reconciling a safety standard with an economic reality the standard did not account for.

2 The Viability Gap

Recall the law of consolidation. A departure is profitable if and only if

$$n > n^* = \frac{c_0}{p - c_v}, \tag{1}$$

where n is the number of units carried, c_0 the fixed cost of the departure, p the price per unit, and c_v the variable cost per unit.

The legal capacity K_L is set by regulation, on safety grounds, and is independent of n^*.

Definition 1 (The viability gap)**.** *The viability gap is*

$$\Delta = n^* - K_L. \tag{2}$$

Proposition 1 (The Viability Gap Proposition)**.** *If $\Delta > 0$, the vehicle is* ***economically unviable at legal capacity****. No level of management, efficiency, or pricing can make the departure profitable while respecting K_L. The operator faces three options: overload, exit, or accept a loss.*

Proof. At $n = K_L < n^*$, contribution is $G(K_L) = K_L(p - c_v) - c_0 < 0$ by the definition of n^*. The inequality is strict, and no choice of p or c_v within the regulatory constraint can reverse it. □

Corollary 1 (Overloading is the only survival strategy)**.** *When $\Delta > 0$, the operator who wishes to remain in service must overload to at least n^*. The minimum overload rate is*

$$o_{\min} = \frac{n^* - K_L}{K_L} = \frac{\Delta}{K_L}. \tag{3}$$

This is the structural cause. The operator does not overload because he is reckless. He overloads because the alternative is to lose money on every trip until he exits the corridor, and the passengers who depend on him lose their transport.

3 The Enforcement Equilibrium

Enforcement is not free, and it is not perfect. Let ρ be the probability of being caught and fined, and F the fine. The operator compares the expected fine to the margin from overloading.

Proposition 2 (The Enforcement Equilibrium)**.** *An operator with a viability gap $\Delta > 0$ will overload if and only if*

$$\rho F < (n_{\max} - K_L)(p - c_v), \tag{4}$$

where $n_{\max}$ is the physical maximum the vehicle can carry. The observed overload rate o^* satisfies

$$\rho F = (o^* K_L - K_L)(p - c_v), \tag{5}$$

so that

$$o^* = 1 + \frac{\rho F}{K_L(p - c_v)}. \tag{6}$$

Proof. The operator overloads to the point where the marginal expected fine equals the marginal margin. Setting the two equal and solving for the overload rate yields the expression. □

Corollary 2 (The fine is a price, not a deterrent)**.** *The observed overload rate is the quantity at which the market clears given the state's enforcement price ρF. The state is not* preventing *overloading; it is* selling *the right to overload at price ρF per unit of margin.*

This is the Becker insight applied to capacity. The fine is a price. The overload rate is the quantity demanded at that price. The state is a monopolist selling overloading rights, and the observed rate is the equilibrium of that market.

4 Why the State Tolerates It

If the state wished to eliminate overloading, it could raise the fine until $\rho F > (n_{\max} - K_L)(p - c_v)$. Why does it not?

Proposition 3 (The Tolerance Proposition)**.** *The state tolerates overloading when the political cost of removing the service exceeds the political cost of the accidents. Formally, the state sets ρF below the deterrence threshold when*

$$W_{service} > W_{accidents}, \tag{7}$$

where W denotes political welfare weights.

The logic is straightforward. If the state deters overloading completely, the operator cannot reach n^* legally and cannot reach it illegally either. The vehicle exits the corridor. The passengers lose their transport. The political cost of that loss is higher than the political cost of the accidents, because the passengers are organised, visible, and numerous, while the accident victims are diffuse and statistical.

So the state sets a fine that tolerates some overloading. The observed rate is the resolution of a genuine trade-off between safety and affordability. It is not a failure of the state; it is the state choosing the lesser of two political costs.

5 The Overload Rate as Shadow Price

This resolution has a precise interpretation.

Definition 2 (The shadow price of the safety standard)**.** *The shadow price of the capacity standard K_L is the fine ρF at which the observed overload rate clears the market. It measures the state's implicit valuation of the safety gain from a unit reduction in overloading.*

Proposition 4 (Shadow price and the viability gap)**.** *The shadow price is increasing in the viability gap Δ and decreasing in the enforcement probability ρ. In corridors where Δ is large — rural routes, poor passengers, old vehicles — the state sets a lower effective price and tolerates a higher overload rate, because the alternative is the loss of service.*

This explains the pattern. Overloading is concentrated in low-income corridors because Δ is largest there. The state tolerates it because the passengers are poorest and the political cost of removing the service is highest. The observed overload rate is not a measure of enforcement failure; it is a measure of the state's implicit willingness to trade safety for affordability in that corridor.

6 The Bucket Interpretation

Overloading is best understood as **capacity inflation** in the bucket system.

A bucket is declared with capacity K_L and a cutoff τ. The operator who overloads is effectively operating a bucket of capacity $K_{\text{eff}} = K_L(1 + o)$ while declaring K_L. The declaration is the legal commitment; the operation is the economic reality.

This is the exact inverse of the kill rule:

Both are violations of the bucket's declared parameters. The kill rule violates the timetable; overloading violates the capacity. The kill rule is usually legal; overloading is usually illegal. But they are the same kind of act: a response to a bucket whose declared parameters do not match its economics.

	Kill rule	**Overloading**
Violation	Refuses to dispatch below λ^*	Refuses to stop above capacity
Direction	Downward	Upward
Protects against	Loss from empty departure	Unviability from legal capacity
Operational form	Waiting at the stage	Extra passengers or tonnes

7 Extension: Axle Load and the Fourth Power Law

The same structure applies to freight, with two important differences: the externality is road damage rather than accident risk, and the damage function is convex.

Proposition 5 (The Fourth Power Law)**.** *Road damage from a vehicle axle scales approximately with the fourth power of the axle load:*

$$D \propto L^4, \tag{8}$$

where L is the axle load. A 20% overload causes approximately $(1.2)^4 = 2.07$ times the road damage of a legal load.

This has two consequences for the overloading equilibrium.

First, the externality is convex in the overload. A 10% overload causes 46% more damage; a 20% overload causes 107% more; a 50% overload causes 406% more. This means the socially optimal fine is also convex — a linear fine is too lenient on large overloads and too strict on small ones.

Corollary 3 (The optimal axle-load fine)**.** *The socially optimal fine for an axle overload is*

$$F^*(o) = \gamma \left[(1+o)^4 - 1\right], \tag{9}$$

where γ is the marginal social cost of road damage at legal load. The fine is zero at $o = 0$ and grows quartically with the overload.

Second, the externality is borne by the state, not the operator. Road damage is repaired by the public road authority. The operator captures the margin from overloading and the state bears the cost of the damage. This is a classic externality, and it is why axle-load limits exist.

But the viability gap applies here too. If n^* (the economically viable load) exceeds the legal axle load, the operator faces the same three options: overload, exit, or accept a loss. The observed overload rate is again the shadow price of the axle-load standard.

Proposition 6 (The axle-load equilibrium)**.** *The observed overload rate for axle loads satisfies*

$$\rho\gamma \left[(1+o^*)^4 - 1\right] = (o^* K_L - K_L)(p - c_v). \tag{10}$$

Because the left side is convex and the right side is linear, the equilibrium may have multiple solutions or none, depending on the enforcement probability ρ and the margin $p - c_v$.

The convexity of the damage function means that the axle-load problem is structurally more severe than the passenger-overloading problem. A small overload is nearly free; a large overload is catastrophically expensive. This is why overloaded trucks are so destructive to roads, and why the fines in most countries are graduated by overload magnitude rather than flat.

8 The Diagnostic

The central practical result of this paper is that overloading is a diagnostic, not a crime.

Proposition 7 (The Overloading Diagnostic)**.** *Persistent overloading in a corridor indicates that $n^* > K_L$ in that corridor. The correct response is not stricter enforcement — which punishes the operator for surviving — but one of three structural interventions:*

1. ***Raise** K_L: larger vehicles, which lowers n^*/K_L and makes the corridor viable at legal capacity.*
2. ***Subsidise the gap**: pay the operator the difference between n^* and K_L, so the departure is viable at legal load.*
3. ***Price the overload explicitly**: legalise the overload and set a fee that reflects the externality, so the operator pays for the safety or road damage he causes.*

Each intervention closes the viability gap by a different mechanism. None of them punishes the operator. All of them are better than the current equilibrium, in which the state tolerates overloading implicitly and the operator bears the risk of arbitrary enforcement.

9 Policy Implications

The overloading analysis yields five policy rules.

1. **Measure n^* before setting K_L.** The legal capacity should be set with reference to the economic threshold, not in ignorance of it. A capacity standard that makes the vehicle unviable is not a safety standard; it is a prohibition on service.
2. **Graduate the fine by overload magnitude.** A linear fine is too lenient on large overloads and too strict on small ones. The optimal fine is proportional to the externality, which is convex in the overload.
3. **Subsidise the gap where it cannot be closed.** In rural corridors, no vehicle size can fill n^* at legal capacity. The state must either subsidise the difference or accept that the corridor has no legal service.
4. **Make the shadow price visible.** Publish the observed overload rate and the implied shadow price. This converts an implicit political trade-off into an explicit one, and makes it possible to debate the trade-off on its merits.
5. **Treat the operator as a symptom, not a cause.** The operator who overloads is responding rationally to a gap the state created. Punishing him without closing the gap does not remove the overloading; it removes the service.

10 Falsifiability

Five observations would refute the theory.

1. **Against the viability gap:** evidence that vehicles overload even when $n^* < K_L$.
2. **Against the enforcement equilibrium:** evidence that overloading does not respond to changes in the fine or enforcement probability.
3. **Against the shadow-price interpretation:** evidence that overload rates are unrelated to corridor income, route density, or vehicle age.

4. **Against the fourth-power extension:** evidence that axle-load overloading does not respond to the convexity of road damage.

5. **Against the diagnostic:** evidence that raising K_L, subsidising the gap, or pricing the overload fails to reduce overloading in corridors where it is persistent.

11 Conclusion

Overloading is not a crime. It is a **diagnostic**.

It tells you that the legal capacity has been set below the economic threshold, that the corridor cannot support the service at legal load, and that the operator who breaks the law is the one who is keeping the service alive. The fine is a price, not a deterrent. The observed overload rate is the shadow price of a safety standard the corridor cannot afford.

The correct response is not stricter enforcement. It is to close the gap — by raising K_L, subsidising the difference, or pricing the overload explicitly. Punishing the operator without closing the gap does not remove the overloading; it removes the service.

> **The theory in one sentence:** overloading is lucrative because the legal capacity has been set below the economic threshold, the fine is a price rather than a deterrent, the externality is borne by the passenger or the road, and the observed overload rate is the shadow price of a safety standard the corridor's economics cannot afford.